

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: MLA0302

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm

converges faster based on the observed results.
(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

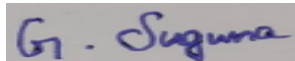
10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

I Suguna G / 19242503 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be "G. Suguna".

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Question 1

Case Given:

A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

Understanding of the Data

The data shows a steady, gradual rise in reward from 10 to 45 across episodes, with no sharp drops or erratic swings. This pattern is the classic signature of a learning curve in Reinforcement Learning (RL): as the agent accumulates experience, the reward obtained per episode moves consistently upward rather than staying flat or fluctuating randomly.

Analysis & Application of RL Concepts

In RL, the agent-environment loop is driven by trial-and-error: the agent selects actions, receives rewards, and updates its value function or policy using methods such as Temporal Difference (TD) learning or the Bellman update. A gradual rise in reward means the agent's estimated action-values (Q-values) are becoming more accurate, and the policy derived from them ($\pi(s) = \operatorname{argmax}_a Q(s,a)$) is progressively selecting better actions. The smoothness of the curve (no large oscillations) further suggests a well-tuned learning rate and a reasonable exploration-exploitation balance — the agent is not overwriting good estimates with noisy updates.

Conclusion

The rising reward trend is strong evidence of positive learning performance and genuine policy improvement. It confirms the agent is converging toward a better, more optimal policy rather than behaving randomly, though full optimality can only be confirmed once the reward curve flattens (plateaus) at its maximum achievable value.

Workflow

Step 1: Initialize the agent's policy/Q-values arbitrarily (or randomly) and set up the environment.



Step 2: At the start of each episode, the agent observes the current state.



Step 3: The agent selects an action using its current policy (exploration vs. exploitation).



Step 4: The environment returns a reward and the next state.



Step 5: The agent updates its Q-value/policy using the Bellman/TD update rule.



Step 6: This cycle repeats across many episodes, and the reward per episode is logged.



Step 7: The logged rewards are plotted; the rising trend (10 → 45) is interpreted as evidence of policy improvement.

Question 2

Case Given:

An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

Understanding of the Data

Three epsilon (ϵ) settings were tested — 0.1, 0.3, and 0.5 — and cumulative reward decreased as ϵ increased, with $\epsilon = 0.1$ producing the best result and $\epsilon = 0.5$ the worst.

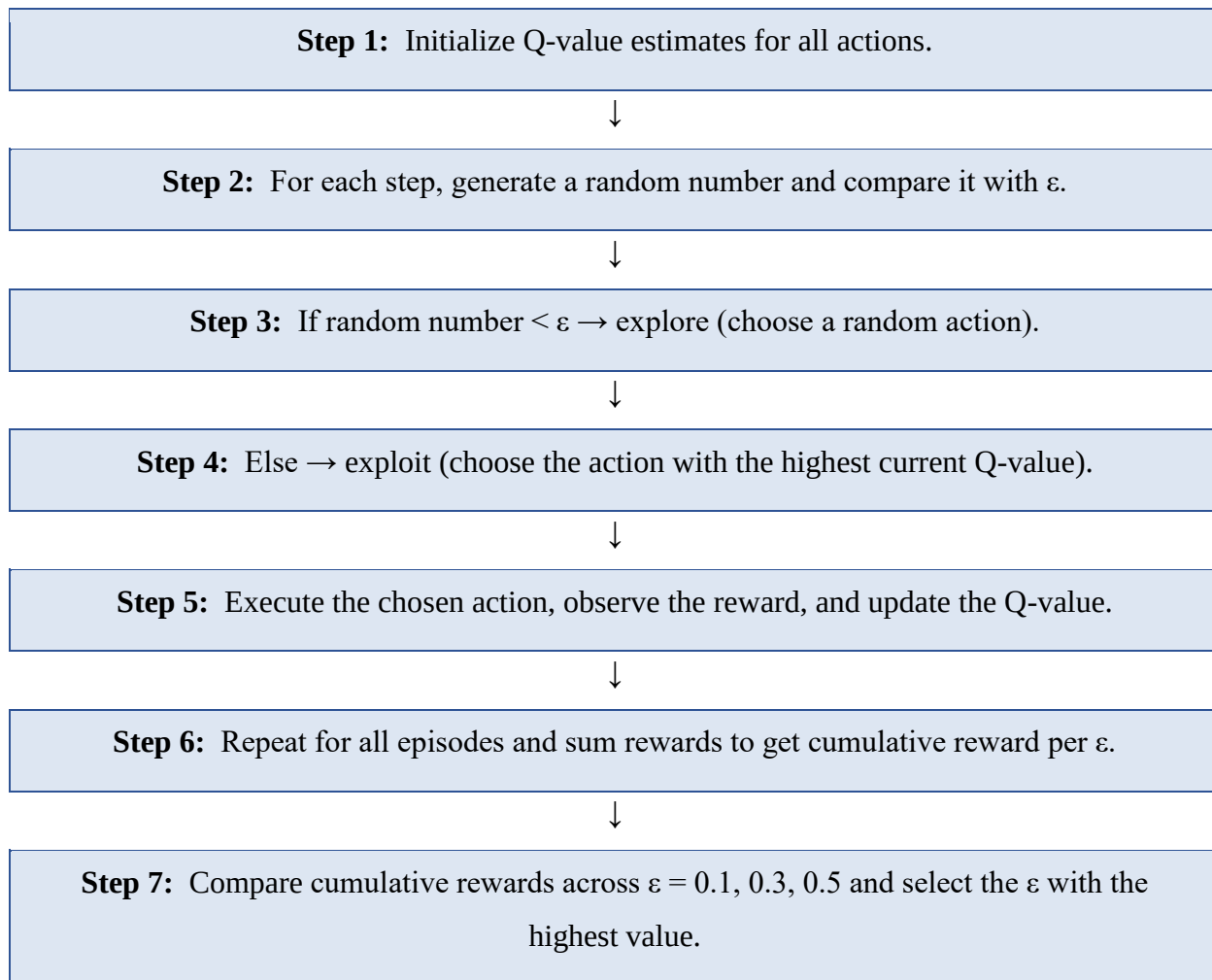
Analysis & Application of RL Concepts

ϵ -greedy action selection chooses a random action with probability ϵ (exploration) and the current best-known action with probability $(1-\epsilon)$ (exploitation). A higher ϵ means the agent more frequently abandons its learned knowledge in favour of random actions, which increases the chance of picking sub-optimal actions and lowers the cumulative reward — this explains why $\epsilon = 0.5$ performs worst. A lower ϵ (0.1) keeps enough exploration to discover good actions early on, but spends most of its time exploiting the best action once value estimates become reliable, which maximizes cumulative reward. This directly illustrates the exploration-exploitation trade-off central to RL: too much exploration wastes reward on random actions, while too little exploration risks getting stuck on a sub-optimal action; $\epsilon = 0.1$ strikes the best balance in this test.

Conclusion

$\epsilon = 0.1$ provides the best performance because it exploits the learned policy most of the time while retaining just enough exploration to keep improving, whereas $\epsilon = 0.5$ over-explores and sacrifices reward.

Workflow



Question 3

Case Given:

Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

Understanding of the Data

Across all three checkpoints (100, 200, 300 episodes), Q-Learning's average reward is consistently a little higher than SARSA's, and the gap remains fairly stable rather than closing over time.

Analysis & Application of RL Concepts

The difference comes from how each algorithm updates its value estimates. Q-Learning is off-policy: it updates using the maximum Q-value of the next state, $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$, effectively learning the value of the optimal (greedy) policy regardless of the exploratory actions actually taken. SARSA is on-policy: it updates using the Q-value of the action actually taken next, $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot Q(s',a') - Q(s,a)]$, so its value estimates are influenced by the exploration noise of its own behaviour policy. Because Q-Learning directly targets the optimal action-values, it tends to learn (and converge toward) higher-value estimates faster, which matches the consistently higher rewards observed. SARSA is more conservative/cautious because it accounts for the exploratory actions it may still take, which typically makes it slightly slower to reach high reward but often safer in risky environments.

Conclusion

Q-Learning converges faster and achieves higher reward in this comparison because its off-policy, greedy target lets it approximate the optimal policy more directly, while SARSA's on-policy nature makes it more cautious and slightly slower.

Workflow

Step 1: Initialize two separate Q-tables — one for Q-Learning, one for SARSA.



Step 2: Run both agents in the same environment for identical episodes.



Step 3: Q-Learning updates its Q-value using the maximum Q-value of the next state (off-policy target).



Step 4: SARSA updates its Q-value using the Q-value of the next action it actually takes (on-policy target).



Step 5: Record the average reward at 100, 200, and 300 episode checkpoints for both.



Step 6: Plot both reward curves on the same graph and compare the gap.



Step 7: Conclude that the algorithm with the faster-rising, consistently higher curve (Q-Learning) converges faster.

Question 4

Case Given:

An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

Understanding of the Data

As the discount factor γ rises from 0.2 to 0.5 to 0.9, the average reward obtained by the agent also rises, showing a positive relationship between γ and overall performance in this task.

Analysis & Application of RL Concepts

The discount factor determines how much weight future rewards carry in the return the agent tries to maximize: $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$. A low γ (0.2) makes the agent myopic — it optimizes almost entirely for immediate reward and ignores the long-term consequences of its actions, which is harmful in tasks where the real payoff (e.g., reaching a goal) is delayed. A high γ (0.9) makes the agent far-sighted, so it values long-term cumulative reward and plans further ahead, leading to policies that sacrifice small short-term gains for larger long-term ones. This explains why average reward increases with γ in this data — the environment evidently rewards long-horizon planning. However, very high γ values can also slow convergence or cause instability in some environments, so the choice should be validated against training stability, not reward alone.

Conclusion

Based on the observed trend, $\gamma = 0.9$ is the most suitable discount factor here, as it best captures long-term reward optimization and yields the highest average reward, provided training remains stable.

Workflow

Step 1: Define the discounted return $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$



Step 2: Train separate agents with $\gamma = 0.2, 0.5$, and 0.9 in the same environment.



Step 3: During training, the Bellman update propagates future rewards scaled by γ into current value estimates.



Step 4: Record the average reward achieved by each agent after training.



Step 5: Plot average reward against γ to visualize the trend.



Step 6: Interpret the upward trend as evidence that long-term planning benefits this task.



Step 7: Select the γ (0.9) that yields the best average reward as the most suitable value.

Question 5

Case Given:

The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

Understanding of the Data

Success rate rises sharply from 45% (at 50 episodes) to 92% (at 200 episodes), a large and consistent improvement over a relatively short training window.

Analysis & Application of RL Concepts

This trend reflects the generalized policy iteration process at the heart of RL: repeated cycles of policy evaluation (estimating how good the current policy is from received rewards) and policy improvement (updating the policy to favour better actions). The steep rise from 45% to 92% shows the agent is rapidly replacing early random/exploratory behaviour with a policy that reliably reaches the goal, which is strong evidence of effective learning. However, 92% is not 100% — the agent has not yet reached a policy that succeeds every single time, so it has not necessarily converged to a fully optimal policy. It may still be exploring, encountering edge-case states it hasn't mastered, or the environment may have inherent stochasticity that caps achievable success rate below 100%. Confirming optimality would require observing the success rate stabilize (plateau) near its maximum over further episodes without further gains.

Conclusion

The improvement indicates strong, genuine policy improvement and near-optimal performance, but the agent should be considered close to optimal rather than definitively optimal until the success-rate curve flattens at a stable maximum.

Workflow

Step 1: Agent starts with a near-random policy (45% success at 50 episodes).



Step 2: Agent interacts with the environment across further episodes, receiving rewards for success/failure.



Step 3: Policy evaluation: the agent estimates the value of states/actions from observed outcomes.



Step 4: Policy improvement: the agent updates its policy to prefer actions with higher estimated value.



Step 5: Success rate is tracked periodically (every N episodes).



Step 6: By 200 episodes, success rate has risen to 92%, reflecting large policy gains.



Step 7: Continue training and monitor whether success rate plateaus, which would confirm convergence to an optimal (or near-optimal) policy.

Question 6

Case Given:

An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.

Understanding of the Data

Among the three learning rates tested (0.01, 0.05, 0.1) on CartPole, 0.05 produces the highest stability, implying the extremes (0.01 and 0.1) each have a drawback.

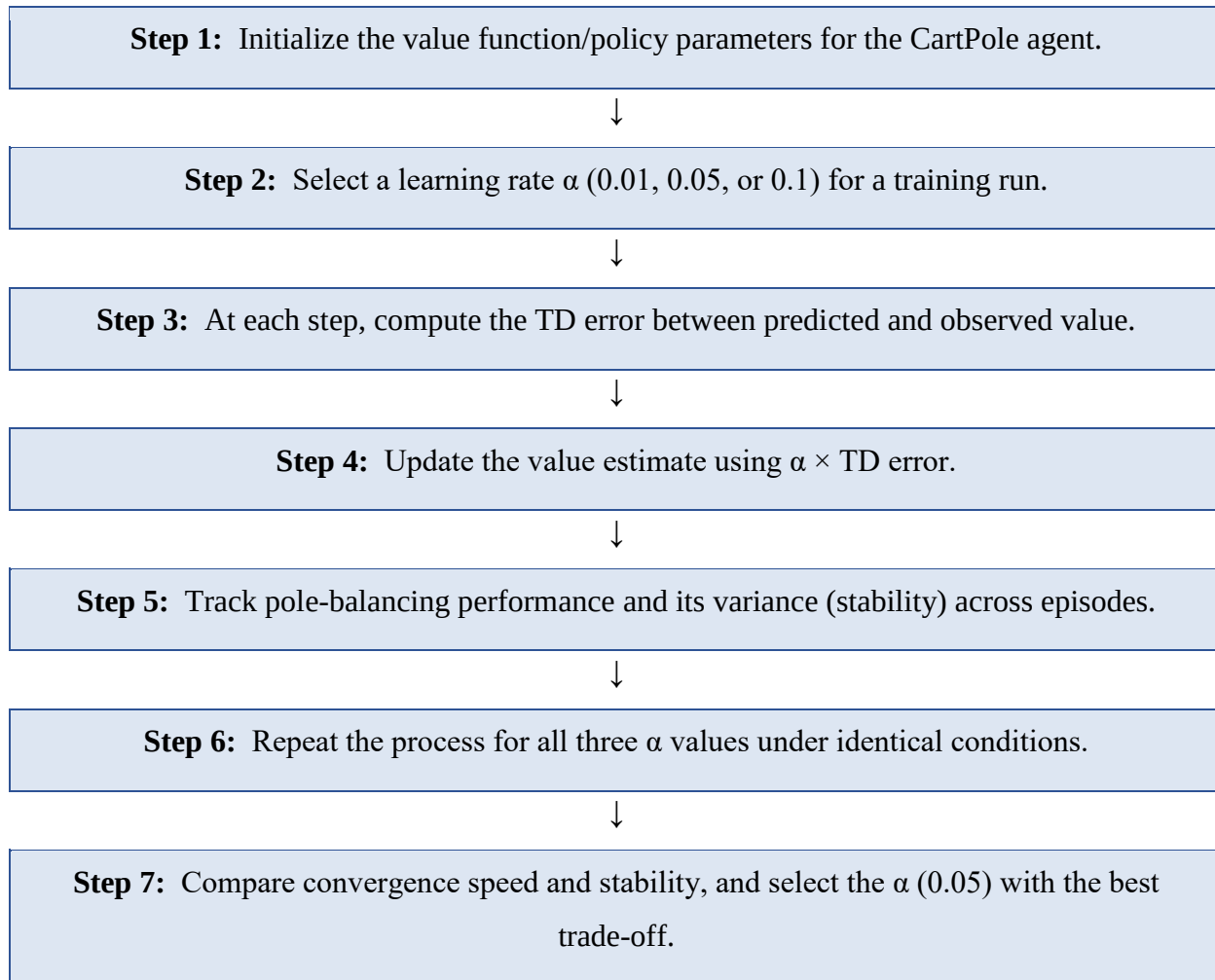
Analysis & Application of RL Concepts

The learning rate α controls how strongly new information overwrites existing value estimates in the update rule $Q(s,a) \leftarrow Q(s,a) + \alpha \cdot [\text{TD error}]$. A very low α (0.01) makes updates cautious and slow — the agent needs many more episodes to converge, even though the training process stays stable, because it barely trusts new experience. A very high α (0.1) makes updates aggressive — each new experience can substantially overwrite prior estimates, which in a dynamically sensitive task like CartPole (where small state changes affect balance) can cause oscillation, overshooting, and instability, sometimes even divergence. A moderate α (0.05) allows the agent to learn quickly from new experience without letting any single noisy update dominate the value function, so it converges faster than 0.01 while staying far more stable than 0.1.

Conclusion

$\alpha = 0.05$ offers the best trade-off between learning speed and stability for CartPole, since it avoids the sluggishness of 0.01 and the instability of 0.1.

Workflow



Question 7

Case Given:

A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.

Understanding of the Data

Three exploration strategies were compared under Q-learning — random exploration, softmax (Boltzmann) exploration, and ϵ -greedy — and ϵ -greedy produced the highest cumulative reward.

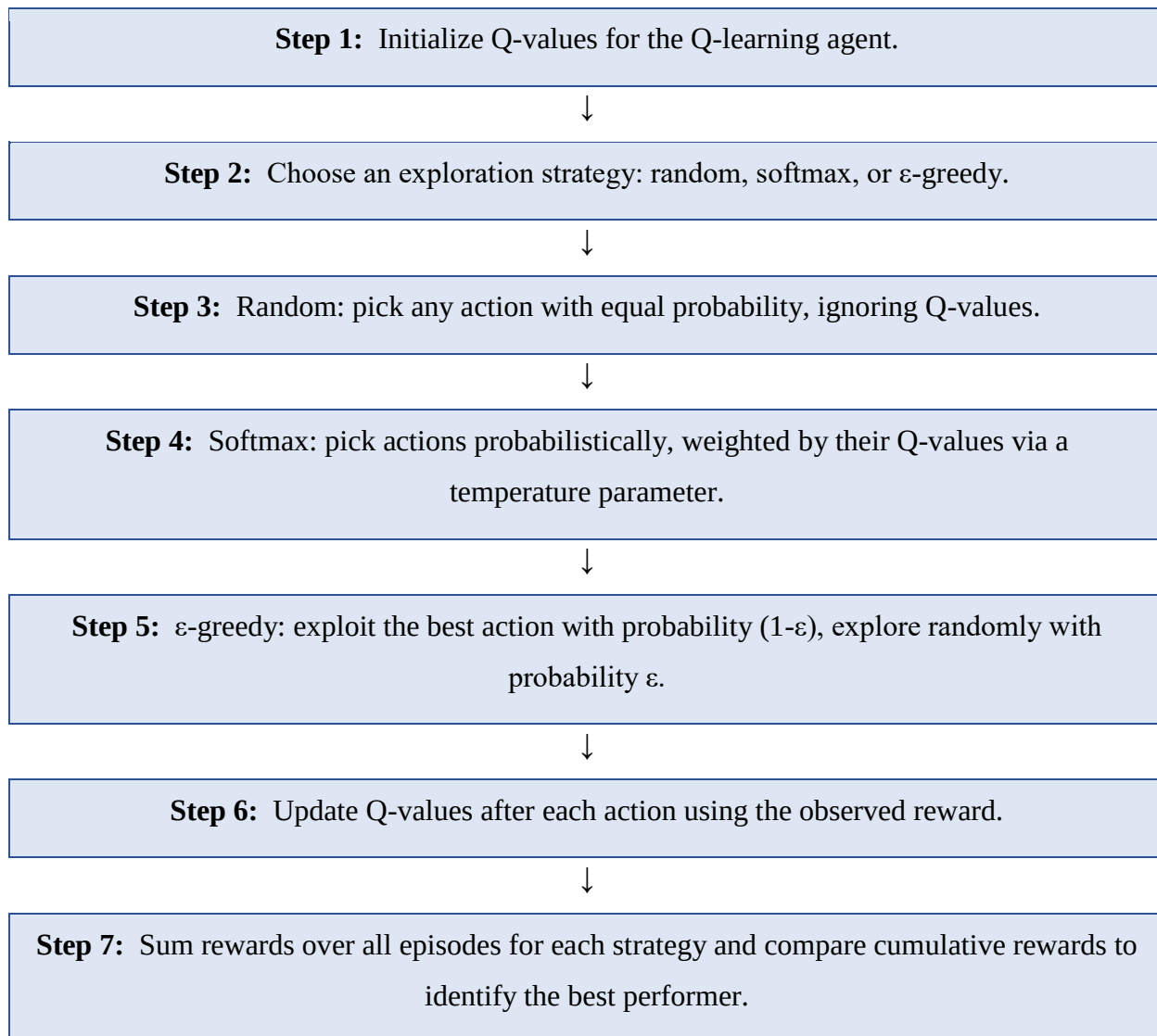
Analysis & Application of RL Concepts

Random exploration never exploits learned knowledge; every action is chosen with equal probability regardless of estimated value, so the agent wastes reward on poor actions indefinitely — this is the weakest strategy for cumulative reward. Softmax exploration chooses actions probabilistically based on their estimated Q-values scaled by a temperature parameter, giving a smoother, value-aware exploration, but it can be slow to commit to the best action when value differences are small, and it requires careful temperature tuning, which can limit performance if not well tuned. ϵ -greedy is simple and decisive: most of the time (probability $1-\epsilon$) it exploits the single best-known action outright, and only occasionally (probability ϵ) explores randomly. This directness allows it to capitalize on learned knowledge more consistently than softmax's probabilistic spread, while still exploring enough to avoid random exploration's total lack of exploitation — explaining its higher cumulative reward in this test.

Conclusion

ϵ -greedy achieves the best performance here because it combines decisive exploitation of the best-known action with just enough controlled exploration, outperforming the untargeted randomness of pure random exploration and the softer, tuning-sensitive selection of softmax.

Workflow



Question 8

Case Given:

An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.

Understanding of the Data

The number of steps needed to reach the goal drops sharply from 50 to 15 after 300 episodes of training in the maze environment.

Analysis & Application of RL Concepts

In maze-solving RL tasks, rewards are typically structured to penalize each step taken (e.g., -1 per step) so that minimizing steps directly maximizes cumulative reward. The steady drop from 50 to 15 steps shows the agent is replacing early exploratory, inefficient wandering with a more direct route, which is a clear sign of policy improvement — the value function is correctly propagating information about which paths lead to the goal fastest. Whether this indicates full convergence to the optimal policy depends on whether 15 steps matches (or is very close to) the actual shortest possible path in that maze: if the true shortest path is, say, 15 steps, then the agent has essentially converged to optimal behaviour; if the true shortest path is shorter, further training may still reduce the step count. Stability of the step count over additional episodes (no further decrease, no fluctuation) would further confirm convergence.

Conclusion

The reduction to 15 steps strongly indicates the agent has learned a much more efficient policy, and if this matches the maze's true shortest path and remains stable across further episodes, it can be considered evidence of convergence to (or very near) the optimal policy.

Workflow

Step 1: Agent begins in the maze with an untrained/random policy (50 steps to goal).



Step 2: Each episode, the agent takes actions and receives a step-penalty reward.



Step 3: Q-values/value function are updated to favour actions leading to the goal faster.



Step 4: Over successive episodes, the agent's path becomes progressively more direct.



Step 5: Steps-to-goal is logged at intervals and shows a decreasing trend, reaching 15 by episode 300.



Step 6: Compare 15 steps against the maze's known shortest possible path length.



Step 7: If steps stabilize at (or near) the shortest path length, confirm convergence to an optimal policy.

Question 9

Case Given:

A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

Understanding of the Data

Testing SARSA with $\gamma = 0.4, 0.7$, and 0.95 shows long-term reward increasing with γ , peaking at $\gamma = 0.95$.

Analysis & Application of RL Concepts

The discount factor governs how strongly future rewards influence the current value estimate; in SARSA's on-policy update, $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot Q(s',a') - Q(s,a)]$, γ scales the contribution of the value of the next state-action pair the agent actually takes. A low γ (0.4) heavily discounts future consequences, so the agent optimizes mostly for immediate reward, potentially missing better long-term outcomes. A high γ (0.95) means the agent gives strong weight to future rewards, encouraging it to plan further ahead and choose actions that pay off over a longer horizon, even under its own exploratory (on-policy) behaviour. Because SARSA already accounts for the actual exploratory next action, a higher γ does not make it reckless — it still respects the risk of its own policy — while still capturing long-term gains, which is why 0.95 yields the best long-term reward without sacrificing SARSA's inherent safety-awareness.

Conclusion

$\gamma = 0.95$ is most effective because it lets the on-policy SARSA agent properly value long-term consequences of its actions, maximizing long-term reward while retaining the safety benefits of on-policy learning.

Workflow

Step 1: Initialize the SARSA agent's Q-values and select a discount factor (0.4, 0.7, or 0.95).



Step 2: At each step, the agent selects the next action using its current (on-policy) exploration strategy.



Step 3: Update $Q(s,a)$ using the reward and the discounted Q-value of the next state-action pair actually taken.



Step 4: Repeat across episodes, allowing the value function to propagate long-term reward information scaled by γ .



Step 5: Measure the long-term cumulative reward achieved for each γ setting.



Step 6: Compare results across 0.4, 0.7, and 0.95.



Step 7: Select $\gamma = 0.95$ as it yields the highest long-term reward.

Question 10

Case Given:

An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

Understanding of the Data

Reward is highly variable during early training but settles into a stable, consistent pattern after around 500 episodes, despite the environment being stochastic (having randomness in transitions and/or rewards).

Analysis & Application of RL Concepts

Early fluctuation is expected in a stochastic environment: the agent's policy is still forming, its value estimates are inaccurate, and it is actively exploring, so both policy immaturity and environmental randomness combine to produce noisy rewards. As training progresses, the agent accumulates enough experience for its value estimates to average out the environment's inherent randomness (analogous to the law of large numbers), and its policy converges toward actions that perform reliably well across the range of stochastic outcomes it might encounter — not just actions that were lucky once. Reward variance shrinking after 500 episodes, therefore, is not just about reaching high reward; it specifically shows the agent has learned a policy that is robust to the environment's randomness, rather than one that is fragile or overfit to a particular sequence of outcomes.

Conclusion

Stability after fluctuation is strong evidence of both policy improvement (the agent now consistently chooses good actions) and robustness (its performance holds up despite environmental stochasticity), which together suggest the agent has learned a reliable, near-optimal policy.

Workflow

Step 1: Agent begins interacting with the stochastic environment using an immature/exploratory policy.



Step 2: Early episodes show high reward variance due to both exploration and environmental randomness.



Step 3: Value estimates are repeatedly updated using observed rewards across many episodes.



Step 4: As more samples are collected, noisy outcomes average out and value estimates stabilize.



Step 5: The policy converges toward actions that perform reliably despite environmental randomness.



Step 6: Reward variance shrinks, and performance stabilizes after roughly 500 episodes.



Step 7: Conclude that stability reflects both policy improvement and robustness of the learned policy.